

ActInf Livestream #034.2 ~ "The free energy principle: it's not about what i

Livestream | 1:42:55

then

[Music]

all right hello everyone welcome to

action flab

live stream number 34.2

it's december 14th 2021

welcome to the actin flab

we are a participatory online lab that

is communicating learning and practicing

applied active inference

you can find us at some of the links

here on this slide

this is a recorded and an archived live

stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome here

and we'll be following good video

etiquette for live streams

if you check out this coda link which i

will post into the youtube live chat

right now you can see a lot of

information on past present and future

live streams and also more information

on if you want to participate because

that's what a participatory lab is if

you're hearing this you have the

affordance to get involved

today in active stream 34.2 we're going

to be having our third discussion on the

paper

free energy principle it's not about

what it takes it's about what took you

there

by axel constant from 2021

and we're just going to be having a fun

dot 2 jumping off

we'll go over any parts of the paper

that we want to look at

return to some of the formalisms and

then also we've written down

a few questions specifically to address

but of course anything people raise or

anything that people ask in the live

chat we can get to that

and

usually it says

next week we will discuss other papers

or we'll be discussing the same paper

but actually it's next year because this

is the last regularly scheduled

weekly journal discussion that we're

doing

and we'll have a little bit of an encore

on this thursday with uh

32.3

when conor hines will join for a dot

three that's kind of like the triple

jump the the hop skip in the jump

uh rather than just the mirror jump

but this is kind of the

ending of a really awesome year of

discussions and we haven't quite set

which papers to discuss next year so if

you have suggestions or you want to get

involved in the dot-coms organizational

unit where we make these kinds of

decisions collectively then just contact

us through any channel

okay

so we'll begin with an introductions and

warm-up section so people can say hello

and they can address any of these

warm-up questions like what they're
excited to talk about today
or just anything else they want to
mention

so this should be fun and thanks to
everybody who's joining on the video
stream and those who are watching live
on youtube

so i'll start i'm daniel i'm a
researcher in california
and

i think today i'm a little bit excited
about talking

about how we got here where we are and
then where we go what is the free energy
principle and active inference say about
where we go

and i'll pass to blue

hi i'm blue i'm an independent research
consultant in new mexico

and i am excited to talk about um
the difference between

living and non-living systems and where
the

fep may come into play there

all right how about dean

sorry

[Music]

i actually don't hear dean

hi sorry about that i'm dean i'm

obviously not fully awake and i'm still
warming up because

i'm in canada in the mountains and it's
snowing as we speak

um what's interesting to me about this
paper and about

its role in terms of fep and active

inference is so once we have identified

the entailment problem then what happens
and i'll pass it over to
stephen
hello steven here
i'm in toronto
and i'm interested in
this
reframing um
i think this paper does i think it's
helpful um
maybe thinking about
you know where was i in the river of
life
when something happened
to take me to where i am rather it sort
of
might be never fully known but the
principles might be useful
i like that rather than
this is the calculation of what happened
which is that temptation so i think
there's something about that going on
here so look forward to pursuing that
more i'm going to pass this over to dave
the two things i'm excited about are
that in this city north of manila which
has rougher the population of pittsburgh
or anaheim or manchester we had one new
case of coven yesterday and zero today
and also as soon as daniel pulled his
pope alexander the sixth routine and
said when will you make an end suddenly
maria and i are just barreling through
the redaction of the
seminar
who
who is there anybody
that's all yeah dave was just mentioning

there the um imminent completion of the
first in symposium transcript
where we're taking the symposium that we
had earlier in the year
and enriching it into something that can
be used for closed captioning as well as
translations and enrichment and
connections to other aspects of our
knowledge base so that will be very fun
and it's also
speaks to how we got here and where
we're going i think because eventually
we'll give that kind of a treatment to
all of these live streams
and
that will be quite interesting
so
steven yes please
i'm actually so because we're using a
slightly different jitsi um let's just
get all of our emojis out of the way if
you go to the hand and then you do
emojis it does like
okay oh perfect disable sounds for all
please disable it
okay now feel free to use emojis you can
signal by repeatedly clicking
how shocked you are for example at what
dean says
okay stephen please
yes
okay
this actually
it works a bit more cleanly as well i
think this this interface um
i was gonna say that i'm interested as
well um
the the question of scale the question

of scale i think there's something in
this is like we're saying um it's not
how you got there well
what scale is you is it scale free is it
scale friendly i know we've been talking
about that but
is is there um
is this is this
you
um
like the royal one
is it like a very broad term or is it
something that's like very much
um at the scale of an agent
um
is it across scales
nested scales
um i'm not saying this necessarily goes
into that but i think because i'm
interested in okay
taking this philosophical
underpinning which at first looks like
it adds an extra burden or becomes a
side issue for a more applied context
actually i think it it does help
um
just back up a bit it helps back up and
in doing that backing up i think it
keeps us with that unifying
principle
um approach rather than a unified theory
temptation
which i think is what often happens
to everyone right we so i think that
keeping it
at that slightly more nascent
understanding may
give us a way to like understand the

types of principles that could be seen
to have played a part which could
therefore be seen to have been useful in
other people trying to do something
similar which is not the same as a
unified theory of
why something
happened

thanks steven dean

so

what the paper brings out for me and and
it kind of ties into what stephen was
just saying is that you
have an entailment question
that you want to have answered
and now

you have to think of that in terms of
what's necessary and what's sufficient
if you look at that

uni-directionally meaning through the
course of time as

the flow or the river or whatever
metaphor you want to apply here if you
look at that one directionally
that's going to kick out one version
and what

this paper points out is that there's
two directions at the same time so
there's a relationship between
meaning and a different form of
entailment

not one causing the other necessarily
but the fact that there are two
and now they are somehow relating to one
another

which is a which is a different animal
and so i think

what i'm going to be really curious

about in terms of the point two of this
of what we're doing here today is
seeing whether or not
we can
address what each one of those things
one
unilaterally unidirectionally through the
course of time and the other strictly
relationally
in terms of how to how do two things
that we know are present
interact with one another i'm going to
be really curious to see what other
people think
might be the difference between those
two very two different views because i
think that's what the author
asks us to entertain is not just
the one
but both
cool i think to to steven's question
about scale i think it's also a
important thing to keep in mind so the
example in the paper which we talked
about in the dot zero and not one was
this bacterium so this is the toy
example that shows that
first off the differences between
the bayesian bacterium and the
variational bayesian free energy
minimizing bacterium
and then shows how it's not simply
sufficient to be minimizing free energy
but rather one has to be minimizing free
energy with the appropriate priors so
it's not enough to be doing just
bayesian statistics
you have to have the right priors in

order to make the right kinds of
decisions

and then the relationship
between free energy minimization and
life is

what is discussed as the entailment
problem so we talked about those earlier
so just go back and check if you'd like
but what is the scale at which the
entailment problem
is even applying

is it the cell membrane of the bacterium
it just aligns perfectly with the
physical scope of the agent

or

is it the colony of bacteria

or is it the bacteria and its niche that
has to be explained how it got there
aren't we always going to have supra
systems

that are going to need their own
explanation

and i think that really speaks to
also what you said stephen about the
difference between the unifying approach
versus the unified theory this is like
past tense

it's been unified what else do we need
to know there's only one

versus unifying is this ongoing process
and it's an approach where we might have
some

guidebooks or tools

or

memes for the road but

it's not over yet we're still
approaching

some other question but then how did we

get that other question
yes stephen and also i don't think that
people need to raise their hand
maybe they can throw up an emoji
silently because it still blinks a
little bit but i think if you just use
for example laughs
then i'll i'll see a flurry of laughter
oh nice
then i'll be laughing as well
that's not probably a bad thing make me
a bit more cheerful um
yeah
i've got this vision of these little
faces now um
there's um
there's something
i'd be interested in a bit more on what
you were just saying there basically
because you you said um
um
there needs to be more
um i'd be interested what people's
thoughts are around
um
what makes this helpful from an applied
perspective
so because one of the big
so i think one of the key things often
in in a more applied perspective is
um there's maybe the application of some
kind of minimal model theory that's been
shown
to
give a type of
transitional
structuring for instance what some
cognitive agent who might be

engaging with the world more like
someone with autism
the nature of how they process sensory
information or other types of situations
so that kind of general
sense of the types of ways
can then be used and maybe applied at a
bigger scale but
generally speaking you also once you get
if you if the applied side of it is
starting to go beyond
a fairly low dimensional model
and into these more complex
um
real life that's not the right term but
applied context
then
these heuristics might be more broad
like the unifying approach is is going
to be maybe more valid and i'm wondering
how does the jump from a biological
philosophy
mindset
that's informing this paper
translate into those so i'd be curious
just what others thoughts are on that
good question how does philosophy
impact or influence application
how do
basic or theoretical biology questions
influence the day-to-day and the way
that different projects are carried out
because
it seems almost
um trivial to say that the projects that
we're implementing whether it's a road
construction or whatever it is
it kind of

has a biological component it's by us
for us
and then it has a philosophical
component even if only implicitly in the
top-down structuring so we're talking
about philosophy and biology i think
that's even what journal the paper is in
so then what does that
mean for how we apply and yeah what else
is part of the approach
i think these are
interesting questions
yeah go for it
one thing i've been very interested in
there's a new field by michael quinn
patton around
principles focused evaluation
which sort of extends his utility
focused evaluation and developmental
evaluation
where you you know your your things are
unfolding so the evaluator is not
it's not evaluating after the fact
but is it part of the strategic journey
to see well what
what sort of um
you could say
in our case what sort of
free energy like minimizing
um
incidents happened that looks
significant to the context
okay so it's it's it's not completely
noisy it's not completely unknown
yet it's complex enough that we're not
following
um a rolled out system that's kind of
relatively dead and relatively um

slapped down on the table each time like
a piece of meat that's dead
it's something that's kind of
evolving and going and i think there's
something quite nice here
to that principles approach um
you know you you and what principles are
going to be useful
when you're looking at understanding
if
there were particular choices in the
particular stream that happened to have
been present
that made free energy minimizing happen
or maybe
when it didn't happen
when the agent didn't manage to
make the
choices and things started to unravel
that could also be interesting because
often it's more interesting to see
you don't really know which one made the
difference when something has been kept
on track
because it it never fully uh you know
spiraled out and magnified but what were
the times when suddenly went wrong
and or
can we find out trace where that
perturbation that started to
rock other parts of the system of uh
interest came from and that that might
be somewhere this type of work
ties in as a thought
daniel can i can i ask a question
without raising my hand
yeah i think we can just go for it in
this one

so

so here's my question and it ties in
directly with this paper

i don't think the author is asking in
the

in the generic sense which came first
the chicken or the egg any more than i
think he's when he's looking at an
entailment problem which came first
sensing or acting which came first
internal or external

which came first um in all aspects of
the philosophical sense of
what's the relationship between
internal external

sensing acting

i think what trips people up and what
gets them on that unidirectional
linear kind of approach is
asking what's first instead of holding
up the relationship between
and then

seeing what falls out the other side

so

if it's an entailment question

is

internal

necessary for external

linearly speaking you could make that
argument but i don't know that that

necessarily holds then if you're saying
does external

then make necessary internal however

if

you look at it as a relationship between
internal and external

both are necessary

so one can be satisfactory for the other

if you're looking at it unidirectionally
but if you're looking at it
bi-directionally now all of a sudden
the frame shifts and
both become necessary
in a relationship in order for the
relationship
to continue forward
so i think for a lot of people
asking which came first the chicken or
the egg from a relational standpoint
doesn't matter
they both have to exist
for the question to continue
not which came first but do both exist
and i think that's the entailment
question that this author is asking us
to
at least contemplate a little bit are we
are we
holding ourselves to one view
that unilateral view when really what we
should be doing is looking at the
relationship and what entails the other
as being either
status satisfying or necessary
great stephen thought on that
and so following on from dean there's
maybe is like you say
when we get into do they exist now we're
into the structural
stuff we're into the structures we're
into the structures where action and
interaction happens as opposed to
a meta system
of egg and you know there's a structure
and i suppose the question can also be
is

do they sufficiently exist
and at what point
so for instance at what point maybe
through
pollution lack of calcium and diet
kinetic
impact
are they no longer sufficiently
persistent
and suddenly one of the eggs can't make
another chicken
okay so that that that could become
you know that then maybe is out of the
how did 50 000 eggs get made
well maybe it's more useful so when
at what point did
point one percent of those eggs
not
continue what was
then outside of
the
pullback attractor or whatever way we
want to say it that was needed to keep
it within
a viable um
a viable living
trajectory so one
kind of funny answer to which came first
the chicken or the egg
from an instrumentalist perspective you
go oh in your sentence chicken
just that's the answer the way you spoke
that question it was chicken and someone
said well what came first the egg or the
ch okay then the egg came first in how
you asked it
so what came first in your model
but that question

smacks of realism

we start thinking about the the actual
way that things might have played out
and is it more the case that there was a
chicken that

that's the realism question

and then when we take this relational
instrumentalist perspective

um there are different kinds of time
relationships falling out and i don't
know if these are the right words but
just a few of them

so

one thing is about the internal and
external that dean mentioned the
partition is simultaneous when we do
a particular partition

of internal external and their
corresponding blanket states
those are all co-instantiated at the
same time in model time or in the
relational instrumental time that
doesn't mean that they had to have been
coming into existence as actual external
entities whether that static structural
ontology or a dynamical process ontology
but once we're in the modeler's time the
partitioning comes into

existence at the same time

and then we have things that are
changing through time

so there's the intra model actin
formalism

like the subscript next to the letter
when we're talking about stay at time
one two three four that's
within the model within the active
formalism changes through time

then there's the
extra formalism like outside of the
kernel of the model which is the
iterated modeling which is also changing
through time because it's not like
any program is just produced and then
that's the final version there's always
an iteration process in the real world
so that's changes through time
and then
somehow spinning that
around
we actually get to this
timeless
and momentary
relationship
that's like the complex entailments
where
we start to
ask about the relationship that's
actually not conditioned on a given
moment but
not sure exactly if that's the right way
to talk about that but there is
something that's it's not time travel
but it's not just forward in time but
it's not merely
reflection or memory on the past
so daniel here's a here's a thing to
think about if
you start with the model
if you begin with the model
you you start it
at
near time zero in even in the model that
axel brings out it's simple then by its
place in the time continuum if you end
with the model

it's got to be much more sophisticated
because a whole bunch of things have
happened a whole bunch of new
information has been available now to
include in the prior
so i don't know that it's it's
complex entailment happens the further
out you wait
for the model to appear if you start
with the model
you're going to get a different result
than if the model is closer to the end
of the loop closure
that's that's what this entailment
question
is actually
surfacing for us
it's when
the model
so where did that come from space time
means when the model when did the model
appear
so that we close one loop and start the
the next update
that's that's where the sufficiency
and the necessity part gets really
strange because it's it's essentially
a space-time question and if we include
two arrows we're now including space and
time
thanks dean stephen
and
within that space and time
i think as daniel has mentioned we can
have these implicit assumptions just in
the way we frame in our questions
which uh would be subliminal but it's
it's sitting there right like you say

by it by even talking about things in
certain ways it privileges a certain
type of thinking and i
i'm really interested um
even in this paper we talk about what
did it take for you to get here okay
so there's also it's almost like that
picture of like
you know the beautiful puppy it's like
and i can imagine asking how did it take
what did it take for you to get here
right but there's also those really
inspiring moments when you see
the really sort of
starved small dog that's kind of in a
really bad shape right and i was
thinking what does it need to get back
to there or something like that i.e a
relatively
viable healthy thriving type of
animal
so that's also and i think that often is
like the priors in some ways when things
have gone out of kilter
there's a need
there's been a prior set of experiences
which have been non-optimal
but hasn't killed the animal
well how do those what priors need to be
fed in to get back on track
although the word track is not the right
term but to get closer to something and
even there i'm struggling i think this
is where the ontology should i be saying
get back to a non-equilibrium steady
state to an attractor to a thriving and
maybe it doesn't need to at that point
maybe there's there's some concept that

doesn't have to be in the free energy
principle exactly it's like some idea of
a thriving type of organism but that
that that
that situation has taken
um has taken the the the
living entity
away from the stream
that would be
desired and has to find a way back
or it won't survive
blue
so
i i was like just to kind of hit on both
what dean and steven were saying like
how did we get here like you i often
think of here as like um like a physical
place but it's also like to this present
time point or here like how did we
become
alive and get to this like
non-equilibrium steady state
so it makes me wonder what dean was
saying like here is also a when right
like so what are all the things that
brought you to this moment
um
and really like
i don't know like so all of the
circumstances that have made you
in this present moment who you are like
whatever you are separated by your
partition
like is the present uh
part of the steady state like like that
that's what makes me think of that i
mean the present can't be like a steady
state system so

it it's because it's like always
constantly like going forward
and
so i don't know it makes me think of
that as an attractor is the present
moment can that be an attractor like is
is that possible stay in the moment stay
in the present when we think about
mental action like like what does that
mean to stay in the moment like can that
be like an attractor state like is the
future an attractive state i don't know
just make me think about that
certainly some kind of
pull from the future
is
what's happening with free energy
minimization
in the physical
gibbs free energy sense
it's as if there's a pull from the
future down the hill but
it's actually this sort of instantaneous
movement
of the reaction
yet it does act as if there's this
long-term pull
and also maybe the necessity and
sufficiency so what does it take to get
here
it depends how
stringent we want to be with what's here
and now because even some tiny tiny
gravitational factor of some other
planet or some
object on the other side of the planet
it's safe to say that some molecule in a
given body would be in a different

orientation

even if trivially so so then is that

going to

mean that everything is

sufficient

like we it was everything that it was

there was sufficient to get us here but

also everything was necessary

is that too stringent or how loosely do

we hold

the system and i think instrumentalism

gives us an answer which is just

how detailed is our model and then

that's the level of stringency which

variables are we going to include

and what precision will we have over

those variables

and then just to the question about

thriving i guess that stephen asks

whether that's in the active ontology or

not so it'll be fun to

find out what is and isn't in the

ontology and then translations or

transpositions

of memes from active to apply to those

other cases so like for example

the person looking at the puppy

and thinking here are the physiological

variables that i can measure there's the

shine on the code and there's the blood

sugar and there's the cortisol level

that's my instrumental model there's

other things happening that i don't know

but those are the ones

and my preference

for the attractor

would be

dot dot dot

and so what policies can i take in
relationship to this animal
so that
its attractor
is now as if it's
here in that three-dimensional space of
shine and blood sugar and cortisol
rather than the current attractor that
it appears to be
orbiting
so then
we can just talk about preferences and
expectations and policies
of the caretaker
and then we've kind of transposed this
sort of like floating obligation
for the puppy to be healthy but then
things do
die
so
we've transposed this kind of
implicit philosophy around a floating
obligation
into a specific relationship
and the policies that a specific
caretaker in their model
are going to engage in
that's uh that's a good point and
i think actually this is when we we find
ourselves getting back into the core of
what's the agent's journey
in different contexts and getting into
that and becomes contextual
and
what's the expectations what's the
surprise
and
when you're into the agent's journey

what was surprising or what's relevant
out of all the
huge amount of surprises that can be
going on across all levels
is kind of what becomes important
potentially
because that becomes where choices were
being made
or not been made
and i think that when we talk about or
when
often there's this question about the
minimal conditions for life
that becomes a sort of false question
because it's again it's in that kind of
the fruit what's the minimal questions
to get life now that's great
if
someone's applied active inference work
is to create life from scratch
i can't see anyone doing that for a long
time
so
however if we want to know
what were the key expectations surprise
or policy shifts
in a living
entity that got them to this new living
entity
because there's changed right i've got
the scars on my hands but maybe i cut
myself with a knife and i had to heal
what were the what were the kind of
journeys on that route and those are the
things outside of a monty python foot
coming down in my head in which case i
don't have much surprise or because i'm
gone

is i'm is what were those
that i think is more interesting and
that is maybe
what this talks about because one of the
key things he talks about is structural
integrity
how to resist
the loss of structural integrity he
doesn't say how to keep the system going
doesn't matter what we see the system to
be
it's if it's acting it's got structural
integrity to be
a thing
with the ability to act
whether that thing is now looked at as a
system
is academic
in the discussion on the metamorphic
systems
where
the paper writes it may be said that the
life cycle that corresponds to the thing
whose integrity is maintained over time
is the process integrity using
bucky fuller's ontology rather than the
structural integrity the structure of
the egg is always being broken down
the larva is always breaking down the
pupa is always the adult is always
breaking down and forming but
there doesn't need to be
structural integrity or the retention of
change at any point
there's a higher order
uh integrity
in the reproductive process
and then that's the question is

okay so can we just look at one nest
mate do we have to look at the colony
colony in the symbionts
colony and the aviotic niche
what is that
life cycle thing
what's the thing's life cycle
is that
gonna be
like a realist
answer where there's kind of one
true
thing
whose
pattern integrity is being retained
or
might there be multiple possible
answers for which thing is having its
life cycle maintained
so then
there wouldn't be
a answer
for
what is life or is are things alive it'd
be model
dependent
and i think that's what we've
one of the things that we've
wrote down to explore
about like what is life obviously
classic questions
and
how
does the fep and actimf
apply to
all kinds of
identified systems
animate meaning

with activity and motion
sentient meaning with sensation
inanimate without
motion
and an incentive without
perception
and
yeah which systems
is it a slam dunk for and where is it
going to be
where is it
overpowered where is it underpowered
what changes would it need to be more
adequate in those different areas
can i add something daniel if i'm a rock
without
anthropomorphizing this if i'm a rock i
can be now and now and now
or i can be then and then and then but
it would seem to me what the free energy
principle tells us is
there is some relationship
to these living things because they
whether they can articulate it or not
they have some sense of the relationship
between
now and then whether then is in the past
and they have a prior or then is in the
future and they have a projection
they're not like
the non-living things which don't
require some relationship between
time points
and i think that's one of the things
that again axel is trying to point to in
terms of an entailment question yes i
can be in the moment and i can be
mindful but the only reason why i'm

aware that i'm in this moment and not
then
in the past or then in the future is
because there is something other than
now
and i think that that again kind of
speaks to
when we can take this and look at it as
bi-directionality as opposed to
unidirectionality
just entropy carrying out what entropy
does
stephen
when
when we think about this free energy
principle applying
and like
dean mentioned you know entropy will be
doing what it does
you mentioned earlier about the
flickering flame okay now i've got a gas
flame okay if i maintain the input of
gas there will be a structural integrity
of the flame over time
so the flame
as a
at one scale with one
perception of what it means to be a
flame it could be seen to be persistent
however
does that mean that it on that is time
average in trying to maintain
if the lens is
a
flame-like structure which maybe if
coupled to our niche somehow we are
trying to make it do that
um

but then becomes the question of
where is the variational free energy
being looked at is it
about
free energy minimization
or is it about
leveraging the fact that there is a free
energy principle happening there is free
energy minimization
there's but there's also just free
energy variation
so i could look at that and i maybe get
some insight by seeing the flickering
flame which isn't structurally hasn't
got that much integrity but it has a
form
that is
always changing yeah has a certain at
some
scale of information
i know i should say scale of information
or something but some qualitative nature
of information persists
over time
where is that flame
in relation to
is it just because there's a
fluctuation that i can extract the
information from the persistence and the
variation even if there's no inherent
motion to reduce
dissipation per se
interesting i think that if we return to
the entailment problem
and the strong and the weak claims
so the strong claim
minimizing free energy is sufficient for
life

the overly generous claim and then the weakness is if the system is currently alive

which as we looked at with that metamorphic slide means that it has persisted in its pattern integrity which I know it was used as structural integrity but that's actually a static concept

it means that it minimized its free energy

so

the flame burning

is kind of the canonical

free energy chemical minimizing system

it's dissipating

literally the chemical potential energy

and it's described perfectly

by the dissipation

of free energy into its environment

and it's even that metaphor of like oh

wait a minute if you put a candle under

a glass jar

it burns out the oxygen and then

no more

same thing happens with

life forms that require oxygen to do

respiration

which basically is just

taking that big drop

in free energy that you can get from

high energy molecules like fats

and then just trickling that

and yes making some heat but also

using it to structure

other

macromolecular systems

internal and external that allow

other fats to get eaten

so

yeah blue

so i want to push back on dean for a
minute um he's like got me stuck in this
like temporal like i don't know zone so
you were talking a little bit about um
the relationship like a rock has now and
iraq has net then

but but i was not clear i want you to
clarify i'm gonna really like push hard
on you i want you to clarify what you
mean

uh like what is the difference there
between a rock and something that's
alive is it a relationship between now
and then be it past or future or is it
awareness

of the past and the future and how do
you know a rock is not aware or that a
bacterium is aware so can you clarify
like what

actually is like this now then

nest

sure okay

so so what i'm saying is i don't need i
don't i don't believe

if if you could i said i don't want
anthro anthropomorphize it but if you
could ask a rock does it need both the
now and then i think the answer it would
give back is i need one or the other i
don't need both

and to the point about the flame

if i if i consider myself metaphorically
to have it a fire burning inside of me
if i'm an engine

i have to know

that in order to keep the fire going i
have to bring in
more energy
i can't just like a candle burn myself
out
so i can't go unilaterally towards the
conclusion i have to continue to bring
up now and then oh i guess i'm getting
closer to the end of my my energy cycle
i have to now
recapitulate
and and bring myself back up to that
place where i can continue to burn
energy so that's what i'm why i'm saying
i think if we want to look at this
from the standpoint of of what
what is actually using now and then
versus
what can continue to exist in it
and i don't know if the word is entropic
but a world of entropy
that's one of those things that we could
look at in terms of what free energy
principle is asking us to
incorporate in terms of a relationship
not just not just the relationship
between two
objects in space but the amount of time
that's incorporated in that relationship
the now
and the then
i don't i'm not i'm not asking us to pin
down what exactly that is i'm saying
let's hold that up as as
as either continuing or not
so
like i think that um
iraq must have a past

in order to exist
in the now
to you
to you and your relationship with that
rock
well even for the rock
to be
itself it didn't just appear out of
nowhere or did it maybe right yeah
that's what i'm saying i don't think it
has a cell because it doesn't have it
then
well so so perhaps
what is necessary
or
what is necessary for something to be
alive
is a delta
between now and then like a rock was a
could have been a rock
then in the past and could be a rock
then in the future
but but doesn't require any kind of
delta between now and then like any kind
of change at all in its structure
although it does change right i mean
like pieces of it fall off and it
crumbles away and turns to dust et
cetera entropy right yeah right right so
i mean it does change over time but but
it doesn't require that that
change and so so perhaps that delta
is one of the things like i need to be
different now
than i was in the past and i need to be
different now than i'm going to be in
the future in order to be alive maybe
here's a a meme it made me think of

eternal sunshine of the spotless mind
it's like a book or whatever
spot it's kind of like a particle
we talk a lot about particular states
so eternal sunshine of the
partic
particle-less mind
like if you don't have a particular
state
which is the internal and the blanket
states those are the particular states
if you don't have the existence of a
partition
it's kind of eternal
it's just like an empty field
now that doesn't have to be a realist
claim about like what was there before
the big bang and or genesis 1
it can just be like in the model
before a partition
and a time scale has been applied
there was no distinctions
but then once you engage a particular
partitioning
then if it is going to persist through
model time and then maybe more
speculatively through real time it does
have to
engage
at least in looking like it's making
adequate policy selection
so one rock we're going to be scratching
our heads over for a long time
but then if we had a field of rocks
some of them would be at and then there
was like um one side of the
rock field had some
corrosive acid and one part didn't like

the rocks that did take the policy to be
on the other side
those would be the ones
that would look like they were
persisting
and so when we kind of zoom out to the
population level which is also what
happens at the end of this paper
when we look at the population level of
a similar type of entities
then we start to see
the strategies that lead to persistence
versus the ones that don't
so
like when people think about the rock i
don't know they rarely talk about other
rocks and about differentiating which
ones are more lifelike or less lifelike
but rather when you only have one
reference point you can't really have a
delta dean
yeah and that that's
this is where i think we can we all of
us that are looking at this free energy
and active inference thing can actually
see a difference between
three rocks on the ground
and a domain
then now and then can be looked at two
ways they were just three rocks on the
ground
and now they're stacked
there's a now and then but that's
different than the partition between
us as living things
and those things that have now be been
reordered and that's what this paper is
asking us to look at do we want to look

only at the way that the rocks have been
reordered or do we want to look at the
now and then of our relationship
to the rock

if we can include both we get a
completely different picture
and of course we even used the rock
example

back in the day back in the dot zero
yeah so here was sort of the two
um life lines
or trajectories because both the person
and the rock

got there there was how
they got there or what it takes to get
there for both

so here is a restatement of the
point of the paper which is our strange
attractor the simple point i make in
this paper is that the free energy
principle is not concerned with the
sufficient conditions of existence
but rather

with what must have been the case
necessity given that you exist
it's not about figuring out what it
takes to be alive

it's about figuring out what took you
there

so here's the lifeline of another system
and um schrodinger's question at least
as it's framed

is kind of like asking you know give me
the binary is it alive or not

but that is

predicated on

us thinking we're alive

i think that's fair to say it's not

stated in there that humans are alive
and that's my reference point or i'm
alive or the writer of this article is
alive but
that
is a very different question so figuring
out what took you there
is a different question than asking
whether the past and the present the
future of the rock
is alive
now
both entities to exist
in a dissipative world
will have done something like minimized
free energy
whether with respect to a complex
boundary condition and a complex policy
space like a human
or
a simpler
boundary condition and potentially a
negligible afford in space
like a rock
like it's almost like
people talk about where the rocks are
alive but then are we alive
and we should have a little bit of an
answer there or what is alive
what scale are we alive at
so can i add to that daniel so if we
look at some lots and lots of those
those figures and we see t
and then we see t plus one and then we
see t plus two we can look at that two
different ways now
those are then
and then and then

in a linear way
or we can look at it as
then as an in relationship to now
and again that's what i think axel is
asking us to contemplate is both
if we look at it simply as then and then
and then and then
i could be a rock
but if i'm now
relating it to some other time point
that's when
whatever whatever else free energy
principle speaks to it says it isn't
just
another point on a continuum
it's in a relationship with something
else
and i think that's what makes it really
interesting in terms of the entanglement
question what's necessary and what's
sufficient
because you can actually look at it as
as another point on a continuum
and it actually kicks out as sufficient
but if you look at it as it a
relationship
to for example now
it becomes necessary and that's the i
think
without
going too far off on the tangent that's
kind of
where it gets really interesting it's
how you are looking at it because
there's a you
asking how did i get here
so i modified the slide a little bit
now the living

red line creature so we're just gonna
say
alive like me
that's the kind of one that matters
in the sense that it has a
quasi-crystalline aperiodic information
structure and it's negantropic those are
the two answers that schrodinger
provides in
1944 what is life
and
we just can't ignore
this
this is where
we're
looking through our lens and our
perspective on the system
so it's not like we get a sort of
um
transparent look
at or an objective or a
monolithic look at the rock it's through
our perception of the past present
future of the rock
and that's where like the specific time
scale that we choose to model it at
because over if you speed up the rock
it's just going to be burning like the
candle
now maybe that takes like millions of
years just in
as the surface reacts with oxygen and
gets oxidized but isn't the candle
oxidizing
and then weren't we just talking about
how we were oxidizing
and then maybe the special part of our
systems

is that we're doing like
this or something
something like that like
we're
back casting
with our metacognitive model
how we got here
well it's been this many years well
since what
since
the hospital signed your birth
certificate
or since you were conceived as a zygote
or conceived as an idea
or what exactly
is
that starting point and i think that's
kind of why the starting point here is
just
off the edge
because again
we may never know what is sufficient and
sufficiency may include everything
but when we talk about necessity
we can actually do
counter factuals we can do loss of
function experiments
necessity can be explored
for complex systems in a way that
sufficiency
may never suffice
blue
so
like you were talking about since you
were
you know born conceived since you were
an idea but what about like since you
were recycled from like previously

existing organic matter like like
i mean literally we are all stardust
right so
like there's
somehow
some kind of partition
gets created
around us like
at conception at birth at when right
like so this whole
temporal
um
thing is is very interesting and it's
like um
you know
would you exist could you exist without
a mother or a father or someone to care
for you as an infant like no um but i
mean and that that distinction is um
very different for different life forms
like bacterium they don't need any care
um they just but but is perhaps like the
reproduction like the act of even binary
vision is that sufficient
care you know like what does that that
relationship um look like in life and in
aliveness and
um you know we talk about like one of
the necessary
um qualities of of life or or you know i
mean i'm not sure we can look at if have
they changed but like the ability to to
reproduce oneself is one of the the
necessary um
qualities for for being alive and um
you know
lots of things we produce that are not
considered alive like i mean lots of

things so um viruses nanobots uh you
know

i don't know it's interesting sorry i'm
like off on a random tangent yeah thanks
blue dean

blue i don't really think that's
that's tangential i think
without without collapsing to like
first year philosophy classes
the fact that we can hold up two things
as important

one of the things that's important is
which came first
we spend a lot of time trying to figure
out order through that lens
the second lens is
there's a there's a here and now and a
then

it doesn't matter which came first it's
just that we understand that the two
things are different
that's the second lens open
they don't they don't play well together
because the the one that wants to figure
out the order is literally trying to
find a source

the other is just saying we're not going
to collapse to one we're going to hold
up two things as long as we possibly can
they're bad playmates we shouldn't put
them on a play date because they're
inevitably going to fight with one
another but if we have both of our eyes
open

what does that say then about us being
able to get a better
grip so that we can let go
when free energy principle is included

or incorporated in these
these sloped moments because i think
it's the two the figuring out which came
first and
here's the relationship between the two
it's not one matters more than the other
that creates the actual slope
that's where as i said i don't want to
turn it into a philosophy course but i
don't think that's a tangent i think
that's what we're trying to reconcile
because we called it attention
we can call it a
you know they don't they don't play well
together whatever but the bottom line is
is that as long as we understand
there are multiple arrows
sometimes we can break them up as then
and then and then and then and then
and sometimes it's about the now and
then
and i think that's the part that for
for most of the stuff that we've seen in
terms of how it's represented it's as
daniel pointed out it's that spontaneity
of partition part
that we have to also include and if we
do
doesn't make it any easier they still
don't get along
but at least it tells us
what's what
what might be at play
that's why i keep coming back to play
it's uncertain who's going to win out in
any particular moment
well i think like while we're discussing
when i think when was the partition

created it is

is important and like you know i mean in

something like binary fission it's

pretty easy to see

you know like when the two you know

cells separate and become you know by a

certain distance by any distance that's

that's when you know the partition is is

made and perhaps that's true for all of

the

the reproducing things or or perhaps not

i mean i don't know like the chicken is

separate from the egg

right going back to the chicken and the

egg but they separate very early so like

the the

you know maybe the separation physical

separation isn't sufficient um like at

that moment you don't have your own

partition because the the baby chick or

the egg could not fully develop without

the mother chicken sitting on it even

though they're physically separated so

so it's this really sticky

situation like where when is the

partition made that makes

someone an individual and that's why i

always go back to that david crack power

paper like when there's a bi-directional

information flow

upward and downward and that that's like

what makes you um

an individual and so like

i don't know that um

an egg a developing egg has

that bi-directional information flow

yet like maybe there's no downward

causation in the egg there's no ability

for the egg to act on its own other than
do anything except develop like the
capability for action or like the the
downward causation like an egg is still
in this emerging it's still going
through emergence but you have to have
both like you know emergence and
downward causation for for individuality
to exist i always go back to that so so
i don't know in a rock is there
downward causation
or emergence like either one of those
things do those exist and so that really
that paper like just nailed it for me
for for what um
you know i don't know that it's
necessarily the answer to schrodinger's
question
but i think um
it does a good job of getting us
somewhere
and i completely agree and and
i guess what i'm suggesting is
entanglement is a feature
not a flaw and and we can enter and in
that entanglement feature set
we can look at
order
right and which came first and we can
look at relation
and so
i know that if if there was an
entanglement even though it's it's it's
multiple light years away that
entanglement tells me what the spin
is and it's not the same direction
there it's it's an opposite it's a
polariness right so

i agree with you
and i think that's the feature here
the feature here isn't we have to we can
only do one or the other
it the reason why we're even speaking to
this is because it has to be both
in order for that slope
to eventually
model out like i'm a big fan of model at
the end i think that's way more
sophisticated because i'm prepared to
let the order and the relationship
hang for a while breathe a bit do some
back and forth
scrap ufc
punch one another and then
i i get my model at the end now i
understand why model at the beginning
but i don't think that's the only way to
go
i think if we can do model at the end
that's
even more sophisticated because we have
now
included more
it's complexity in play
let's return to now
v3
of our
figure
so
these the colors are subject to change
all right but red is realism
blue is instrumentalism our favorite
statistical instrument is
the actin
process theory which is a corollary of
more axiomatic

more general less applied free energy
principle

and then this is another system and the
system of interest in this paper was a
bacterium

so

whether were structural realists or
pretty much any stance other than the
most

hardcore denialism

we exist like if we're not alive then

what is the conversation to be had

but when we ask how did i get here which

is really happening or when we look at

another system that's also really

happening those are real events

but both the reflexive question

and the

exteroceptive

scientific modeling question

are going to get filtered through

instrumentalism

i don't know if there's some way out of

that

it's kind of like the difference between

saying well you can escape plato's cave

but then can you escape like using your

eyes

or using your generative model it's like

a whole nother

level of cave

and then

we have

like

active and the free energy principle

this classic figure

organism and the environment

and then that is shown in the case of a

brain
as well as in a bacillus
that particular partitioning
about how we can think about the
perception cognition in action and
impact of autonomous systems
once you're past
the veil
you're all within your instrumental
model
so that's where the question like well
what were those priors
in my bayesian bacterium and then the
future is unknown because we can say
well in my model the bacteria is
minimizing free energy or i'm talking
about this bacterium and i think
physical systems minimize free energy
it's a physical system so i think it's
going to minimize free energy but as we
established
in this paper
free energy minimization isn't
sufficient to be alive it's not
sufficient to persist because the candle
is minimizing free energy too
so we don't know just because it has
existed and is minimizing free energy
that it will exist in the future so
that's kind of the unknown
and then once we're
post-partitioned
we're all in the instrumentalism world
so
even the framing of what is life
it's an implicit realist
playing field
versus how do we model life

and then

the answers that schrodinger provides

look for the information rich according

to your information theory quasi crystal

structure

or look for using your thermometers and

your heat sensors

look for systems that appear to be

locally creating order but then creating

disorder surrounding them

i don't know if anyone really denies

that those are good ways to study life

or that those are

heuristics that can be applied it's just

like saying well your instrumentalism

doesn't align with my realism

okay

but what is your realism and who told

you about it and how did you find out

about it what are you gonna do about it

what instrument are you going to use to

investigate it

so

it's interesting how the paper

shines a light

on how the entailment problem is really

linked to some of the biggest themes

that we talked about this year

with realism and instrumentalism

because again the

entailment problem runs through and

through

what do you think dean

yeah i would say

that nails it because if you want to

look at it as

which is more important realism or

instrumentalism fine but if you're

looking at the relationship
between
realism and instrumentalism and how they
relate to
that's going to make a fuller picture it
doesn't necessarily make it a cleaner
simpler picture in fact it makes a
messier more complex picture
oh well sorry
there's a relationship between those two
things get over it grow up
time to grow a nice long white beard and
and smile because guess what it ain't
going away just because it's more
complicated
so i have a problem with like
i don't know realism overall because
like what is even real
like i
view the brain as just an instrument
through which we you know act and infer
like this is this is it and every living
system in my mind has like this
um you know kind of computational input
output that they use but but it's an
instrument that they use and like the
idea of realism is very um icky to me
because
it's so subjective like what is even
real i just don't i can't buy into the
notion of some kind of shared reality
for anyone we all have different
affordances different preferences like i
don't even know if i see the same color
when i see yellow that daniel sees or
dean sees like well you know how i
perceive yellow can be totally different
from how you perceive yellow and and

that's why we all have different colors
like favorite colors right with
different colors that we like more and
like less
so you know i mean and people are color
blind and so we know that it's just not
that way so
i don't know
like it's all instrumentalism in my
like
world like we're instruments
i don't know
i think it returns to the unifying
approach versus unified theory
we've been really
mired in the
mix-up of realism and instrumentalism
and it sometimes is very
slippery
like how things are
but then that's the twist with the
theory of everything
and all the free energy principle for a
particular physics and all of that it
uses
sometimes even some of the same
terms like thing
but then
by thing within the act of ontology
we might be referring
to this
particular thing
so the blanket and the internal states
and this particular thing is a cognitive
model of
and then could we have a model of those
kinds of things
maybe

but could we have a model of
the
real things or would this always be
a sort of blurry discussion because
everybody would be
talking about their model anyway and
it's also a lot like um
the layers that existed
in 32
which we'll return to in two days
where there was the lorentz system that's
like the realist
system now it's one that we defined but
it's really that way the time evolution
of the lorentz system truly is defined by
those equations that were granted to it
so that's sort of the realism claim and
then there was a first and a second
order approximation to it
which are also
things but they got cleaner and cleaner
and more tractable
as we instrumentalized
the thing we were looking at
and then
the question is always well how much of
the attributes
of the real system
does the instrument retain
do we retain
false positives like kind of things
about the territory that end up not
helping our map
do we have false negatives like we
simplified the map a ton but we ended up
missing
really important features of the
territory

and if the territories are complex as we
suspect they are to have persisted this
way in a negentropic world
well if they're complex
then
how will we
thread the needle
with the appropriate coarse grainings
as not to include false positives and
false negatives in our map
because just having the full description
of the system is not going to be useful
and then
anything simpler
we're going to be at super risk of
dropping features that matter
and so 32 showed that it was possible
just at the mathematical level for
example to make an approximatable
laplacian
approximation so tractable laplacian
approximation
but still retain
the lyapunov exponent characteristic of
a chaotic system
so we we
abstracted or we made approximations too
but we still retained one of the
essential features of
and so
maybe
it's going to be harder to do that to a
bacterium than a lorenz system because
the reality of the loren system is just
numbers whereas the reality of bacteria
is something probably different
but it's a similar kind of question
and

maybe we have some of the early versions
of the tools that we would at least want
to have in the future dean
yeah and i'm not defending either
realism or instrumentalism but i think
in this paper and that paper which we're
going to go back to
if we if we use real or realism
what they both point to is we can get it
really right
or we can get it really wrong or some
degree of that i think that's where
realism
as a as a concept comes in it's not an
absolute
it's not a truth
it's the variable
as opposed to the instrument the
instruments is trying to find out the
precision
and reality is about imprecision
so again it's the relationship between
those two things i'm not looking at
realism in the classic definition of how
other people see it
i'm comparing it
to the instrument which is trying to
gain the precise
and then so so what's the not precise
okay i'll just i'll just throw that
under realism for now that's how i
square it
so so dean we're on the same page then
because like accuracy is relevant
yeah exactly
exactly so i i don't think you and i are
apart or even remotely
you know distant

but it's how you see that relationship
if you if you frame it one way something
kicks out and if you frame it a
different way something else kicks out
and i'm not going to tell people who
frame it the other way that they're
wrong
they're relatively
safe in in their frame but it's got to
be two frames
at minimum otherwise
we're we're lost
we're a stone
per se
we're just going along for the ride
and i don't know that that's
i don't know that that's fair to stones
well i don't know that we have any other
choice but to go along for the ride
honestly i mean
like
and this goes back to you know the
discussions that i've had about free
will and um i discussed this with mike
levin and i i mean really like you know
if if we are just the sum of our
generative model like if that if we
that's just us we have no choice but to
act accordingly
unfortunately or or fortunately for us
blue how's your i just the confusion's
built in sorry i just wanted to say i
think the confusion is built in i think
the uncertainty part of this is built in
we know that we're eventually our demise
is near we just don't know when
right and that's why i keep coming back
to the where where did how did i get

here is essentially asking
when
and that goes back to way back when we
were talking about people with um
alzheimer's and in the hippocampus
integrating where and when
yeah and the the grid cells and the
mirror neurons and
all kinds of ideas
that people have frames
that are
instrumental
takes
that blur the line because they're about
real systems but then
we can
bring a little bit of an actin flavor to
it
so
just
i have one note then a question
so here
here's the red living system now it's
shrunk and turned into an image
and then here's here's the orange person
studying bacteria
so now it's like the blue and the orange
person
they can be having a conversation
through the scientific literature or on
a live stream
or whatever like this is where we start
to go from that
rock is it alive or not to here's a
population of rocks
which ones are
succeeding at being lifelike versus
failing in terms of their dissipation

so now here we have
a minimum two people looking at the
bacillus each with their own
autoreflexive timelines and maybe the
orange person
doesn't like active
that's fine they're using some other
framework but they're not breaking out
of the instrumentalism
trap
they're just doing some other
instrument
and then they're looking at the bacteria
through their lens
and then there's a conversation to be
had
but it's almost like the more layers the
more super systems you add here
it starts to be like
i don't know less and less about the
bacteria
and oh well in your culture you were
raised to think this way about this kind
of scale or this kind of system and
this is an interesting conversation
but it's still nucleated around that
bacteria and the model
and then blue my question for you was
over 21
how has your
feelings on free will changed
um
so like you know after i i'd say like
the real the real um
like mindset change shift happens more
like at 25. um at least for me
but um i don't know like you know i
think that the whole experience of being

a teenager is like identity seeking and
like

putting yourself under the illusion that
you have free will or you have like such
a desire to have free will that that
there's um

i i mean like that's what it is to be a
teenager is like you go through this
like identity crisis and then you
liberate from the parents right
and so i think that there's that like i
definitely experienced that if that's
what you're asking

she already underwent the phase
transition

yeah

yeah i mean the phase like i already

yeah i think um

you know as teenagers we have like this
illusion of and desire for freedom
that is um

like

what we're supposed to have and feel at
that at that phase in life
in order to break free and and maybe the
age is irrelevant i mean maybe in some
ancient prehistoric cultures they felt
that at 10.

but the

the point is is that the desire to you
know

um

is it physical separation or or to
solidify

your own partition right like um the
desire to be your own independent person
so i think like i i
thought that you know freedom and free

will
existed at that point and now like um
i mean i feel like we're
essentially enslaved by our generative
model we're enslaved by the past and the
future in the now
i think that takes us to one of the last
prepared
questions we had
one of the key words of the paper was
historical science
and
we talked in the dot zero
about how
there's all these historical sciences
you won't see
chemistry biology
physics not those
stem
sciences
but history
and then all these other
sciences
that draw data from records of past
events
as opposed to experimental or
operational science but asked it then
and asked it now aren't experiments in
the past
i don't see how
merely having hypothesized and planned
and then
acquired data
when you're analyzing it it becomes in
the past
so somebody who's studying heraldry
could hypothesize oh in this library on
this shelf in this book i might find

something important and i'll analyze it
then
and then that data is in their past
so
the historical sciences are
contrasted with
maybe what we could call the
future sciences so not like the future
of science which we hope is awesome and
participatory and accessible and whatnot
but science is about the future
which range from
the sort of speculative
and almost metaphysical
to the extremely
applied
where do we sit now
with historical and anticipatory or
future oriented sciences in the now
dean
so i'd like to try and take a a first
stab at that
let's and let's use it let's use the
free will question
if we looked at it historically as
bring out the scrolls this is what
happened or we look into the future and
say
oh
this this person's thought a lot about
what's going to happen tomorrow let's
build the scroll for tomorrow
that's one way of looking at the
question of what is free and will
if we look at as a relationship we could
look at it as
free will or
follow will meaning i can't break free

i'm i'm entrapped or whatever
and then what we could do is we look at
the relationship
as an ongoing
an ongoing exercise between free and
follow now neither
neither the free or follow are going to
fall neatly and tidally into a scroll
but they have a relationship
whether we scroll it out
as a historical thing or we scroll it
into as a projecting thing
but the ability to do both not just
examine free will as a linear
question
but also
be able to relate it to so what's
what is non-free will it's like
will follow
and then now compare the follow and the
free
give it kicks out a different model
the one that's one that's messier
one that's up for debate
and i think putting putting both of them
up on the shelf and examining both is to
look at
philosophy
essentially
so
i think that i i recently just learned
that futurology is a science
um which which was super fascinating to
me um
like the study of of predictive methods
right
um
and there's so many ways that that we

you know have used to predict but um
something that i thought was was really
really interesting was um
they
used to before the internet was like
trolled by bots you used to be able to
take like message boards and chat boards
and like you know essentially twitter
streams um and i'm not sure how accurate
this is anymore because there's so much
like just artificially generated text
but when it was humans creating the text
they used to actually
do
text mining as as like a study of the
future
um
like so like a prediction analysis based
on text
analysis so it's um and was very
effective so i hear but i think that
this like study of future predicting be
it oracles or astrology or what have you
is really kind of fascinating because
you can
use all kinds of things active inference
for prediction or anything right so um
i don't know i think it's interesting to
compare and contrast the different ways
and um
the ways that
if you're open to accepting insight from
tarot cards or the iching or the oracle
of delphi
or
active inference or
you know text analysis if you're open i
think that

all of the future
or potential potential future avenues
can can be very insightful like
that's a possibility if only opening you
up to the fact that that's a possibility
i think that that that's the the real
value in in
futurology thanks blue dean
yeah i think you're i think you're right
i think the law of large numbers
and the fact that we now have the
ability to look at much larger data sets
and and apply
probability and statistics to those
things should give us
some sense of direction that we probably
didn't have affordances to before
and the messiness and this is why a lot
of times philosophers and scientists are
in that same tension of relating to is
because
that still doesn't collapse down to
i go and see my my tarot card reader
and he gives me an answer and i believe
it
because that's not a law of large
numbers so now do i follow or am i free
i don't know
it's messy right so yeah anyway well and
does history repeat itself right like
there's the old adage history repeats
itself can we use history oh
can we use history to um
predict the future
right
that's where it's sort of mobius strips
strange loops around
when you make the model based upon the

past
that's anticipating the future
and then you start acting that way going
forward in the future
it's almost like the timeline has folded
because now the past is
influencing the future
in a in a way that
might start to diverge
from not like what it would have done
obviously any policy is going to diverge
from some counterfactual
but
it starts to um potentially
lose its
model adequacy
because
events from the past
whose context has changed
like let's just say the correlation
between
two cryptos
we look at the past correlation but the
context is changing
so we're gonna have a model that's fit
on their correlation in the past
and then especially action on that but
even without our action on that
that relationship may
change
and i think that's something interesting
about actin that it it
puts perception cognition and action
in the same loop
always unfolding kind of like uda
so that there always is the opportunity
for flexibility
as surprises come in

and then also what you were saying about
like

free will following well freeing
following

and then to bring it back to realism and
instrumentalism like

here's the free will module in my model
it's like here's how it works here's
what it does here's the inputs here's
the outputs so does the thing have free
will yep in my model it does

what about the actual thing
i thought i told you

we stopped talking about that a few
years ago or something like that like
we're not talking about that or
that's an interesting question but
now we're at square zero with what is
real

instead of i made a hypothesis about
free will module that's just what i'm
calling it that's not a free will module
that's a random number generator or
that's something that is doing its own
prediction and then makes a decision
based upon something else or that's like
a preference vector

okay

dean

yeah and again today

i don't i don't think it matters i know
most people would say it does but i
don't think it matters whether you start
with a slide that has a blue silhouette
looking at a gray rock
or you start with a slide
that seems appears to be much more
more um that's what's the word for it

much more further along in terms of
orange lines
you can go both directions yay
that's the thing that we should be
celebrating
like it may appear from a from a
constructivist
point of view which came first
that there's no way that you could have
started with the orange and ended up
with the much
much less
um
informational 10 but yes you could
and we oftentimes do
that's that's actually if you go from 13
to 10 in the slide deck that is free
energy
playing out yeah we could look at this
one and we could go oh that's minimizing
we're
we're talking about instrumentalism so
we just know that when we draw an arrow
of some agentic system we know that
we're talking about instrumentalism
and we know that the instrument that
we're talking about is free energy
principle active inference so that's how
you step back from here to here
and then just go okay so and then we
know that models have their own internal
time and that that's not the same as
like real time so let's just simplify
that
and then let's just
a little bit less academic not worried
about which paper i should write let's
just focus on the motivation of that

paper is it alive it's like
we step back
now
i don't know is that
it's like backwards and forwards in
kairos
as we move forward in kronos dean
and that's where
instructionalism and i forget what that
it was a hippolyta paper i think yeah
interactive construction
performance where the action is
yeah
that's where this that's where those
four slides prove out her paper
or at all paper because the reality is
it's not
simply constructivism it's
disentanglement
there's there's a product at the end of
it that appears to be a construction
don't get me wrong
however what actually generates four
versions four iterations is a dis
entanglement not an integration
and we have to be able to hold up both
if we're actually going to get to a
place where we understand house
understanding
emerges
and that's where i that's where i find
this stuff really exciting because if
it's about interaction here's the proof
let's in the last few minutes
where do we go
so not where we came from
[Laughter]
or when do we go

um mom are we there then

um

blue will be working on some fun

literature analysis the history of

active inference literature and free

energy principle half 22.

we hope to

see many new papers

coming in

from

authors that we know as well as from new

authors and new perspectives

and selecting

cutting edge papers and recent papers

that are coming out and preprints ones

that might not be done see where the

live stream can

insert itself into that literature

stigma g

maybe go back to some earlier

classics

for contrast and understanding how ideas

change through time and where we are now

who we are now all these other variants

of the question that we talked about

i mean

we're gonna be doing it on wednesdays

uh next year

starting one hour later

at uh i think 16 utc at 8 am pacific

so

for

the weekly paper discussions or just any

other system of interest

where are we going or what will we do

differently in 22

i'm curious to see we're going to have

our 40 second or 40 second live

livestream that's gonna be epic we have
to like choose wisely choose choose a
paper wisely and also like our first
paper of the year is gonna maybe set the
tone for what we're gonna you know learn
and do and be about this year what was
our first paper last year in 21 daniel
it was um

adam saffron and colleague 13.1 let's
look on the coda site
um oh we haven't updated it yet but it
was 13. on the cybernetic big five
oh that was epic too that was really
good and actually like you know the big
five was that was my first time learning
about it and it was kind of
we covered all of those this year uh
several times so we'll have to have to
similarly choose like an
all-encompassing
first paper

hopefully we go into more participation
and
more expansion and
more

broad acceptance or recognition of the
fep i hope that's where we go
that would be cool

do we have any other
comments or we can give sort of a yes
dean please

one last thing

i think

i didn't come in i wasn't there a year
ago

but one of the things that has recently
happened is that we've introduced
another walnut shell the potential for a

0.3

which i actually think for most people
would be like yeah okay so what you're
gonna you're gonna stare at your navel
longer but i actually think it's a
really exciting

all right pete dean yeah

um

dean

please from the from the oh wow

all right dean yeah but the doctor

affordance is cool so that's a really

good point dean um the walnut shell the

navel gazing the dot three

continue

yeah i just was gonna say i think it's

it's actually a really exciting time

because when we add another shell we're

actually practicing the active inference

instead of just

talking to it in different versions

and that that to me is if next year

means that there's

more

reveal more

q

unknown

and we actually have to play with that

uncertainty space as opposed to walking

into it and saying well we've now had

this many live streams so

now we could no we should actually be

doing more shells

1p

and more options

hidden

that's when it gets really exciting so

that's what i'm looking forward to so

that's a super good point dean and so if
people have other live streams that they
wish that they were a part of or you
know other authors want to come back for
a revisit to do a dot 3 of any live
stream that they've been on before like
reach out
we always have the affordance to make an
anything.³

um
so yeah like we can revisit old things
and move on to new things and um do both
of them together history the weird
folding right

yeah we can do guest streams all kinds
of fun stuff and then also just so
people know

um if you're ever part of the live
stream you'll see this slide
asking you to go through a checklist
this is our 21 version

and we hope people go through it i think
sometimes they do

lots of information on participation
on things to check before you start
if you're a participant so not like the
author of the paper

um but just somebody who's
gonna be hanging out and
giving any thoughts on a paper
there's some checklist for you for a
presenter like an author who's gonna be
presenting a talk

there's information for them
there's also the broadcast

role
so that's the person with a stable
internet and a computer that can run a

streaming software
there's a conversational
facilitator yeah exactly
dean's cabin
dot stream um a little stream through a
tin can
then there's the organizers checklist
and then we're always
growing and changing as active inference
lab because
we're a participatory lab and that
includes every single possible role
including many that are not listed here
so
if it's something that you're interested
in and if you're listening and your
regime of attention has extended this
far
then
maybe you'd like to join a live stream
or make a suggestion
or play one of these other roles because
it is a multiplayer game
and i hope we're having fun when we get
to do it so
dean and blue and steven from earlier
and dave from earlier as well today
always appreciated it was an awesome
year i think we all
learned a lot by doing and updated our
generative models
thanks daniel really appreciate it
thanks blue
fantasy